

# Bibliography

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# Chapter 1

## Background and Motivation

### 1.1 Problem Statement

#### 1.1.1 Fuzzing Background

Fuzzing is nowadays a widely used approach for software testing and vulnerability assessment. Its main strength derives from the fact that it could rapidly produce various inputs to be fed into a target executable. In general a fuzzer can be divided into the **Input Generator**, that is responsible of providing inputs to the **Executor** that execute them. During the execution the **Defect Monitor** checks if the input executed ends in a crash of the target. [?] Since the input space, in real cases, is infinite, it is impossible to test the application on every single input, so the need of a precise input selection and mutation arises, in order to discover as many edge cases as possible.

Depending on the availability of the source code various strategies could be applied in order to guide input mutation.